Antenna Radiation Efficiency Measurement in a Reverberation Chamber

GRADUATION PROJECT
BACHELOR OF PHYSICAL SCIENCE

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Overview

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2. Basics of Antennas and Radiation
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5. Measurement Methods in RC
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10. Conclusion

Introduction

Antennas and Chamber Measurements: Goals and Applications

- Antenna Parameters Characterization:
  - **Anechoic Chamber** (AC): Classical way - Time consuming - Costly.
  - **Reverberation Chamber** (RC): New method (statistical proprieties) - Faster and economical.
- Miniature Antennas $\rightarrow$ *Body implants* [1]
- *Constraint* on Radiation Efficiency (η_{rad}):

Radiation Efficiency Trade off [3]

$$\eta_{\mathrm{rad}} \propto \frac{\mathrm{Size}}{\Delta f}$$

Basics of Antennas and Radiation

Antennas: Role, Evolution and Measurements

- First developed by Heinrich Hertz in 1888 (Maxwell's theory: 1862) [10].
- Devices allowing to **transmit** and **receive** signals over great distances.
- Play a crucial role in a lot of scientific domains (Figure . 2).
- Important **intrinsic parameters** to estimate in a RC or AC *depending* on the application (D,Δf,η, η_{rad}, ...).



Figure 2: ILS Localizer Antennas [7]

Basics of Antennas and Radiation

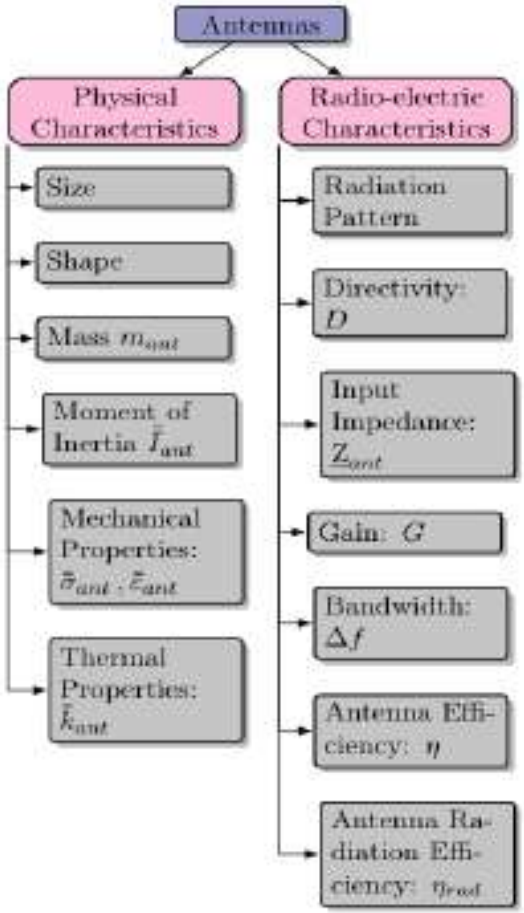


Figure 3: General Characteristics of Antennas [9]

Antennas: Definition, Intrinsic Parameters and Radiation Principle

Definition

An **antenna** can be defined as a *usually metallic device* used to convert a **wire-guided time-varying electrical signal** by a conductor into an **electromagnetic signal radiated in free space** and vice versa.

Physical Radiation Principle:

Larmor Formula [11]

$$P_{\text{rad}} = \frac{q^2 \| \vec{a} \|_2^2}{6\pi\varepsilon_0 c^3}$$

Basics of Antennas and Radiation

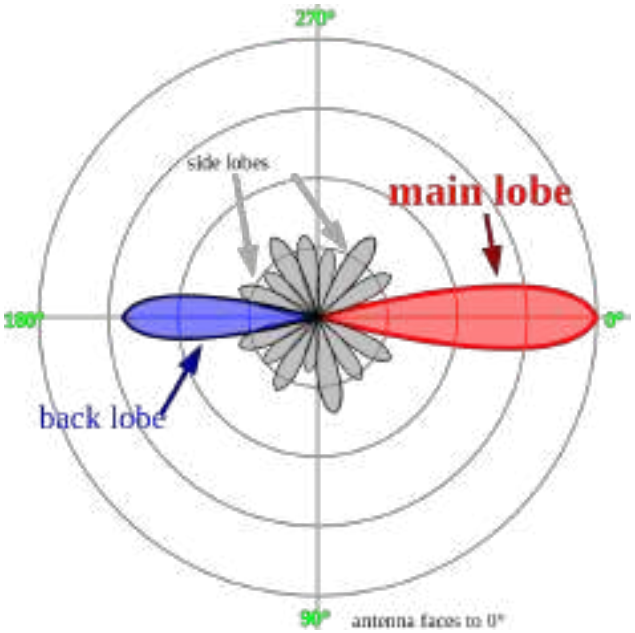


Figure 4: 2D Polar Radiation Pattern $D = D(\theta)$ of a Directive Antenna [12]

Most Important Antennas Parameters [2]:

- Radiation Pattern: 2D/3D plot of various **radiation properties** of the antenna (FIGURE . 4).

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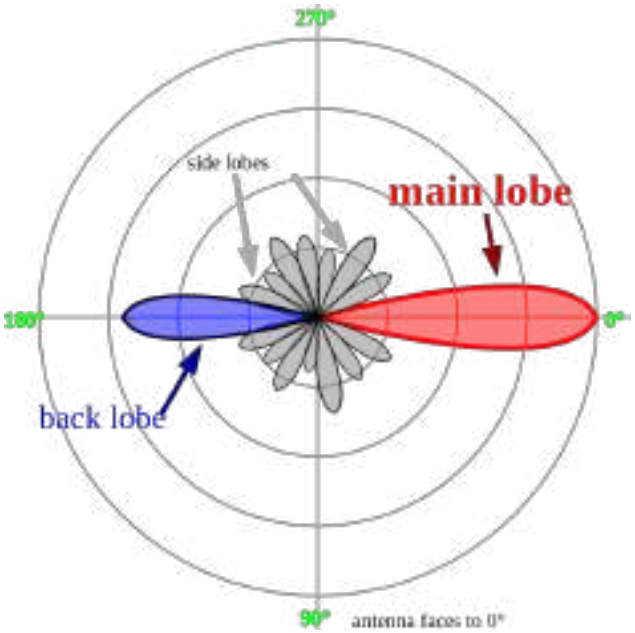


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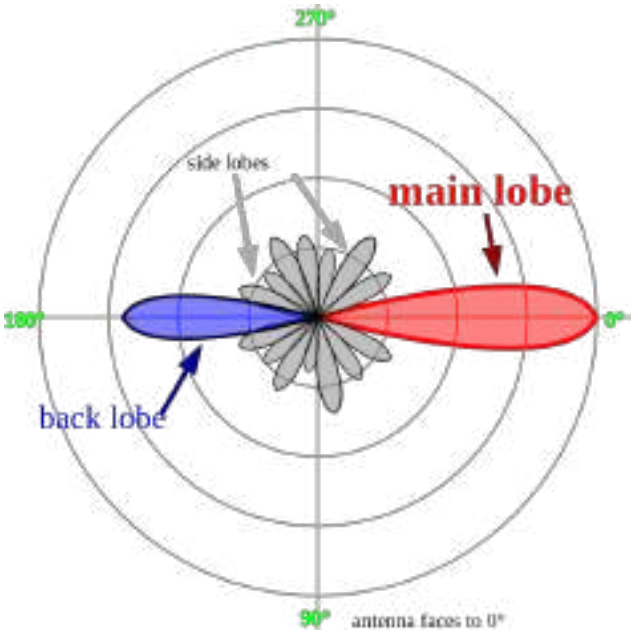


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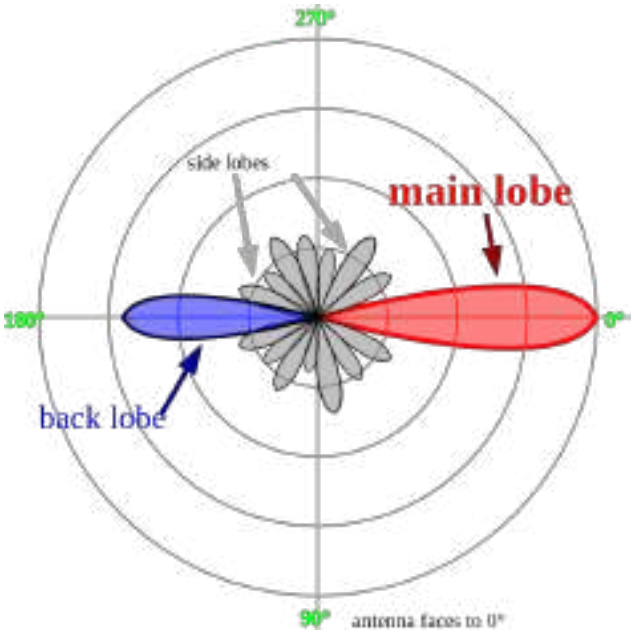


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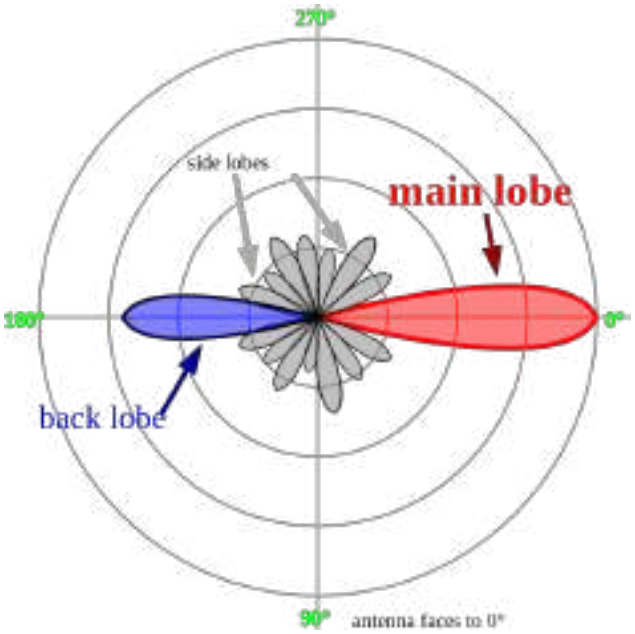


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- Bandwidth Δf: *Region of the frequency domain* the antenna is the **most efficient** to radiate or receive energy.
- Gain G:

$$\mathrm{d}G(\theta, \phi) \equiv 4\pi \frac{\mathrm{d}U^{\dagger}(\theta, \phi)}{P_{\text{ant}}} = \eta_{\text{rad}} \mathrm{d}D(\theta, \phi)$$

Antennas: Circuit, Efficiencies and Power Transfer

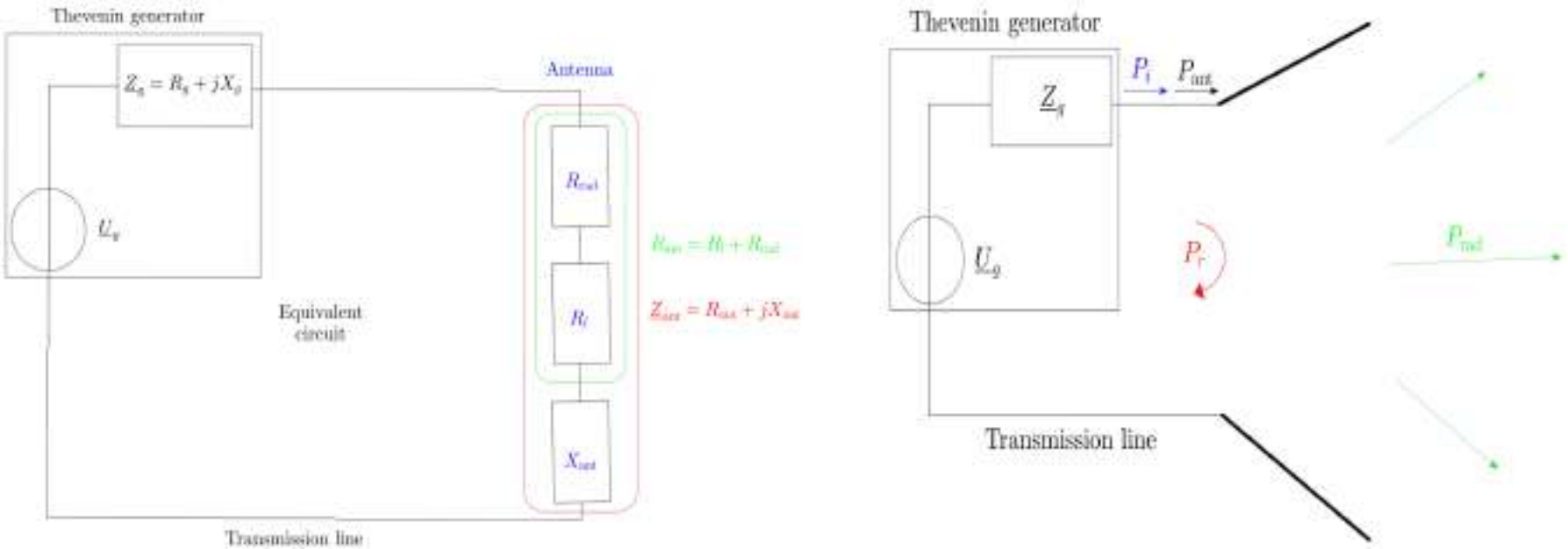


Figure 5: Antenna: Basic Circuit

Basics of Antennas and Radiation

- An antenna is not an *ideal* structure (ohmic/dielectric losses): $\eta = \eta_r \times \underbrace{\eta_d \eta_c}_{\eta_{\mathrm{rad}}}$

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Power radiated into free space

$$P_{\text{rad}} = P_{\text{ant}} \eta_{\text{rad}} = \left(1 - |\underline{\Gamma}(f)|^2\right) \eta_{\text{rad}} P_i$$

- Reflection coefficient:

$$\underline{\Gamma}(f) = \frac{\underline{Z}_{\text{ant}}(f) - \underline{Z}_g^*(f)}{\underline{Z}_g(f) + \underline{Z}_{\text{ant}}(f)} \qquad \underline{Z}_g^*(f) : \text{Complex conjugate of source impedance}$$

S-Parameters

S-Parameters: Definition and proprieties [4]

- High frequencies (MHz-GHz): Complete characterization of electrical devices by Scattering parameters $[\underline{b}] = [\underline{S}][\underline{a}]$.

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Definition and Power waves

$$\underline{S}_{ij}(f) = \left(\frac{\underline{b}_i}{\underline{a}_j}\right)\Big|_{\underline{a}_i=0} \qquad \underline{a} = \frac{\underline{U}_n^i}{\sqrt{\underline{Z}_g}} \qquad \underline{b} = \frac{\underline{U}_n^r}{\sqrt{\underline{Z}_g}}$$

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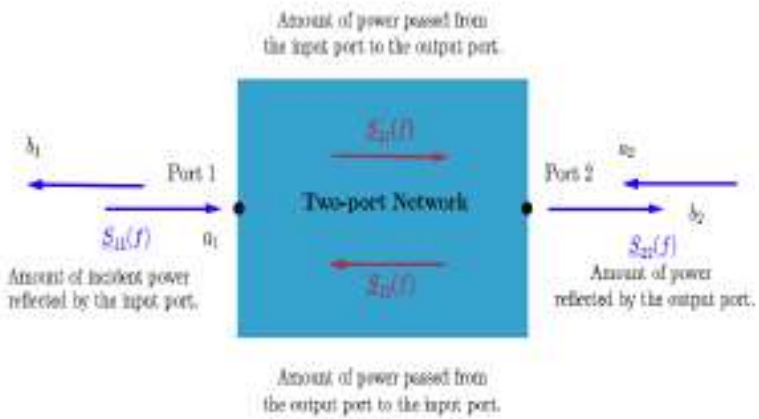
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- Subscripts:
  - i: Port where the energy **emerges** (*output port*).
  - j: Port where the energy **enters** (*input port*).

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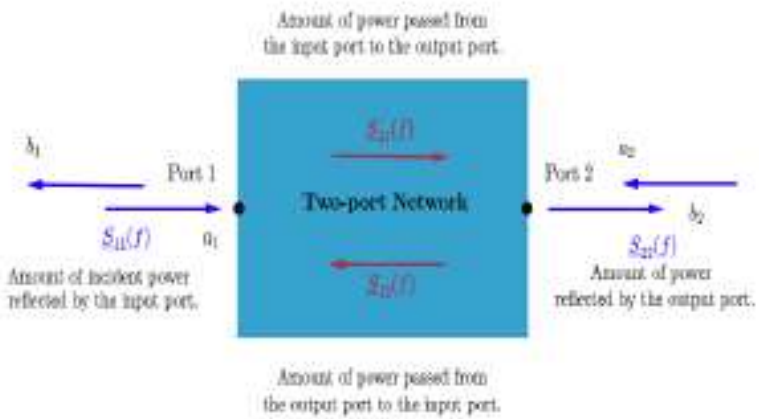


- *2n-linear-pole*: Characterized by a square $n \times n$ *scattering matrix* [S] with n^2 scattering coefficients ($n = \#$ *of ports*)

Figure 6: Two-port Network: S-parameters [7]

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- Two antennas exchanging RF energy can be considered as a two-port network (FIGURE . 6).

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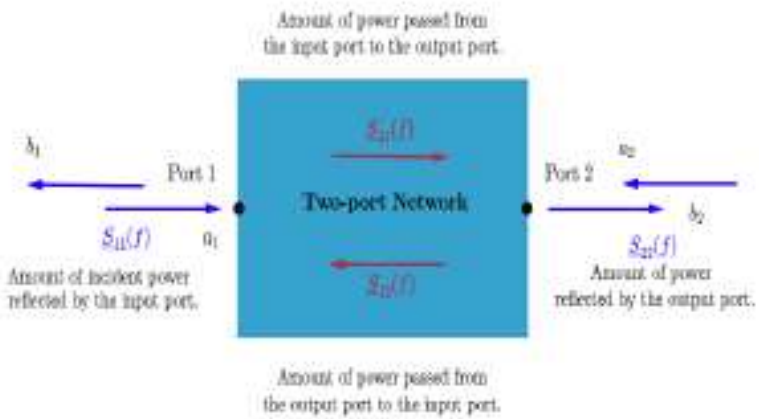


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- Two antennas exchanging RF energy can be considered as a two-port network (FIGURE . 6).
- $\left|\underline{S}_{ij}(f)\right|^2$ is dimensionless (ratio of power) and $[|\underline{a}|] = [|\underline{b}|] = \sqrt{\text{Power}}$.

Basics of Reverberation Chamber

Reverberation Chamber: Definition and Use

- Parallelepipedic cavity with a mechanical stirrer

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Reverberation Chamber: Definition and Use

- Parallelepipedic cavity with a mechanical stirrer
- Over-sized cavity, thus electrically large and increases wave reflection
- Statistically homogeneous and isotropic environment $\rightarrow$ Statistically uniform fields

Basics of Reverberation Chamber [5]

Reverberation Chamber: Electromagnetic field

Random distribution of the field $\rightarrow \langle \vec{E}(\vec{r}, t) \rangle = \langle \vec{B}(\vec{r}, t) \rangle = \vec{0}$

Basics of Reverberation Chamber [5]

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$$\begin{cases} \langle \| \vec{E}(\vec{r}, t) \|_2^2 \rangle = E_0^2 \\ \langle \| \vec{B}(\vec{r}, t) \|_2^2 \rangle = \frac{E_0^2}{Z_0^2} \text{ with } Z_0 = \sqrt{\frac{\mu}{\varepsilon}} \end{cases}$$

Basics of Reverberation Chamber [5]

Reverberation Chamber: Q-factor and Power

- Increasing of the amount of reflections with the Q-factor

Definition: Q-factor

$$Q \equiv \frac{\omega U}{P_d}$$

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$$E_0^2 = \frac{Q P_{\text{rad}}}{\varepsilon V \omega}$$

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- Average Received Power:

$$\langle P_{\text{rec}} \rangle \equiv \langle |\underline{I}|^2 \rangle R_{\text{rad}}$$

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- Increasing of Received Power with the Q-factor

$$\langle P_{\mathrm{rec}} \rangle = \frac{Q\lambda^3}{16\pi^2 V} P_{\mathrm{rad}}$$

- Composite Q-factor from different power losses:
  - Wall losses $\rightarrow$ Predominant
  - Absorption of loading objects
  - Aperture leakage
  - Loads of receiving antennas

$$Q^{-1} = Q_1^{-1} + Q_2^{-1} + Q_3^{-1} + Q_4^{-1}$$

Measurement Methods in RC [8]

Radiation Efficiency Estimation from Reflection Coefficient Measurements: Free Space Contribution and Backscattering Response

In the RC, two types of contribution for S-Parameters

$$\underline{S}_{xx}(f) = \underbrace{\underline{S}_{xx,\text{Free Space}}(f)}_{\langle \underline{S}_{xx,}(f) \rangle} + \underline{S}_{xx,\text{Stirred}}(f)$$

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- Backscattering: $\underline{S}_{xx}(f) = \underbrace{\langle \underline{S}_{xx}(f) \rangle}_{\underline{S}_{xx,\text{ Free Space}}(f)} + \left(1 - |\langle \underline{S}_{xx}(f) \rangle|^2\right) \eta_x \underline{H}(f)$

Measurement Methods in RC [8]

Radiation Efficiency Estimation from Reflection Coefficient Measurements: Q-factor estimation

- Average transfer function:

$$\langle |\underline{H}(f)|^2 \rangle = \frac{\langle P_{\mathrm{recId}} \rangle}{P_{\mathrm{rad}}} \Rightarrow \langle |\underline{H}(f)|^2 \rangle = \frac{Q_x}{Z_0 \varepsilon V \omega} \frac{\lambda^2}{4\pi}$$

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- Expression of the Q-factor:

$$Q_x = \langle |\underline{\mathrm{S}}_{xx}(f) - \langle \underline{\mathrm{S}}_{xx}(f) \rangle|^2 \rangle \frac{2 Z_0 \varepsilon V \omega}{\lambda^3 \left(1 - |\langle \underline{\mathrm{S}}_{xx}(f) \rangle|^2\right)^2 \eta_x^2} = Q_x^{\#} \times \frac{1}{\eta_x^2}$$

Measurement Methods in RC [8]

Radiation Efficiency Estimation from Reflection Coefficient Measurements: Radiation Efficiency calculation

- Intrinsic to the chamber:

$$\frac{Q_1^{\#}}{\eta_1^2} = \frac{Q_2^{\#}}{\eta_2^2} \iff \eta_1 = \eta_2 \sqrt{\frac{Q_1^{\#}}{Q_2^{\#}}}$$

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- Final expression:

Radiation Efficiency

$$\eta_1 = \sqrt{\frac{\langle |\underline{S}_{11}(f) - \langle \underline{S}_{11}(f) \rangle|^2 \rangle}{\langle |\underline{S}_{22}(f) - \langle \underline{S}_{22}(f) \rangle|^2 \rangle}} \left| \frac{1 - |\langle \underline{S}_{22}(f) \rangle|^2}{1 - |\langle \underline{S}_{11}(f) \rangle|^2} \right| \eta_2$$

Experimental Protocol: Electromagnetic Field Stirring and Measurements



(a) Reverberation Chamber at ESYCOM Laboratory



(b) RC's Mechanical Stirrer at ESYCOM Laboratory (rotation step of 5°)

Figure 7: *ESYCOM* Laboratory: Reverberation Chamber and its Mechanical Stirrer

Experimental Protocol: Electromagnetic Field Stirring and Measurements



(a) Log-Periodic Antenna: REF Antenna ($x = 2$)



(b) Miniature Patch Antenna: AUT ($x = 1$)

Experimental Protocol: Electromagnetic Field Stirring and Measurements

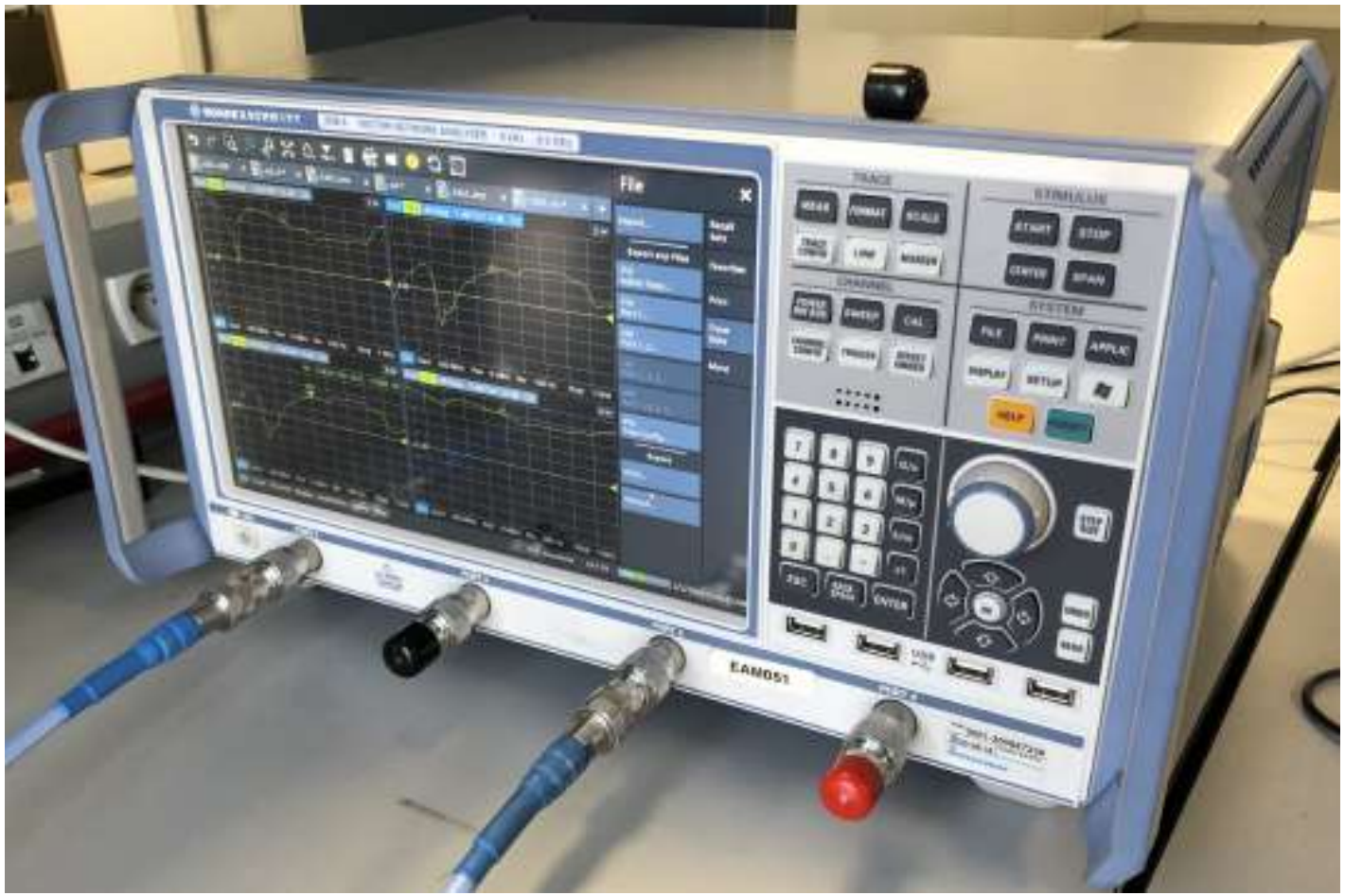


Figure 9: Vector Network Analyzer (VNA)

Data Organization

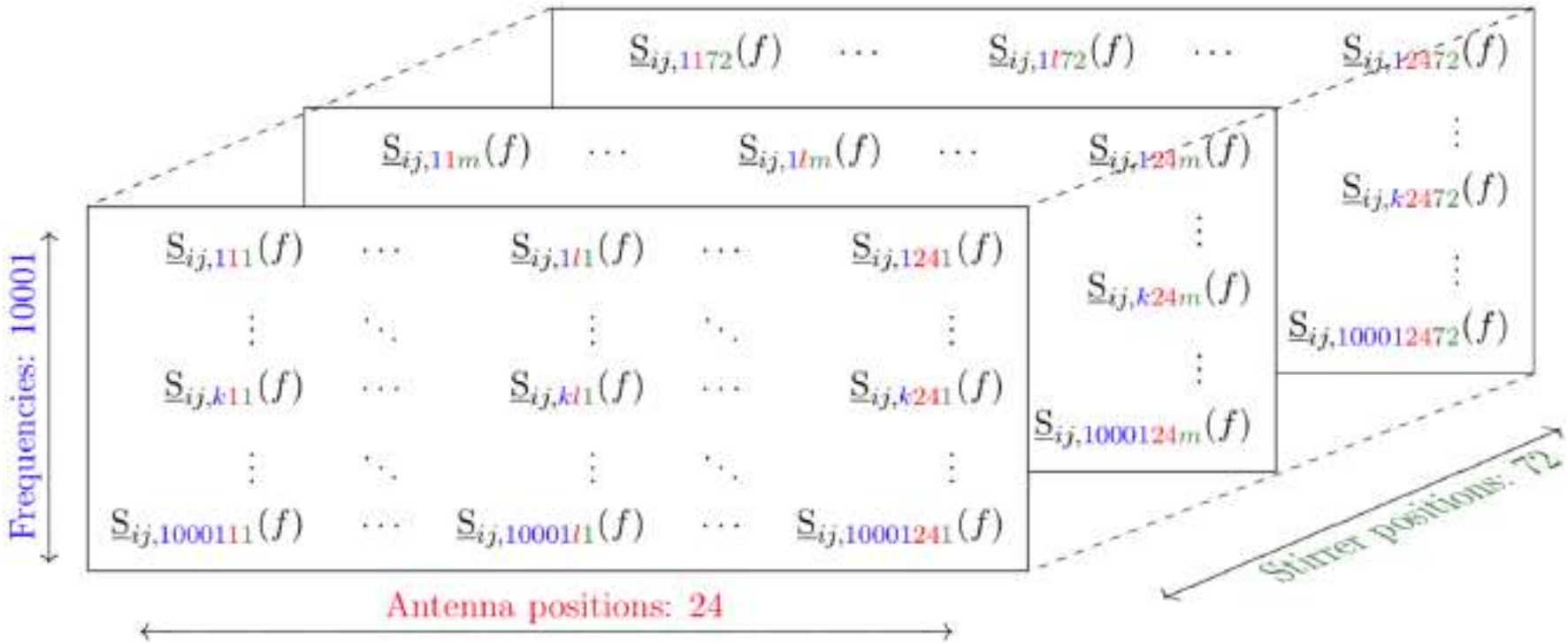


Figure 10: $\mathbb{S}_{ij}$ 3D Matrix: Data Organization on Matlab

Data Processing and Antenna Radiation Efficiency Establishment (Besnier)

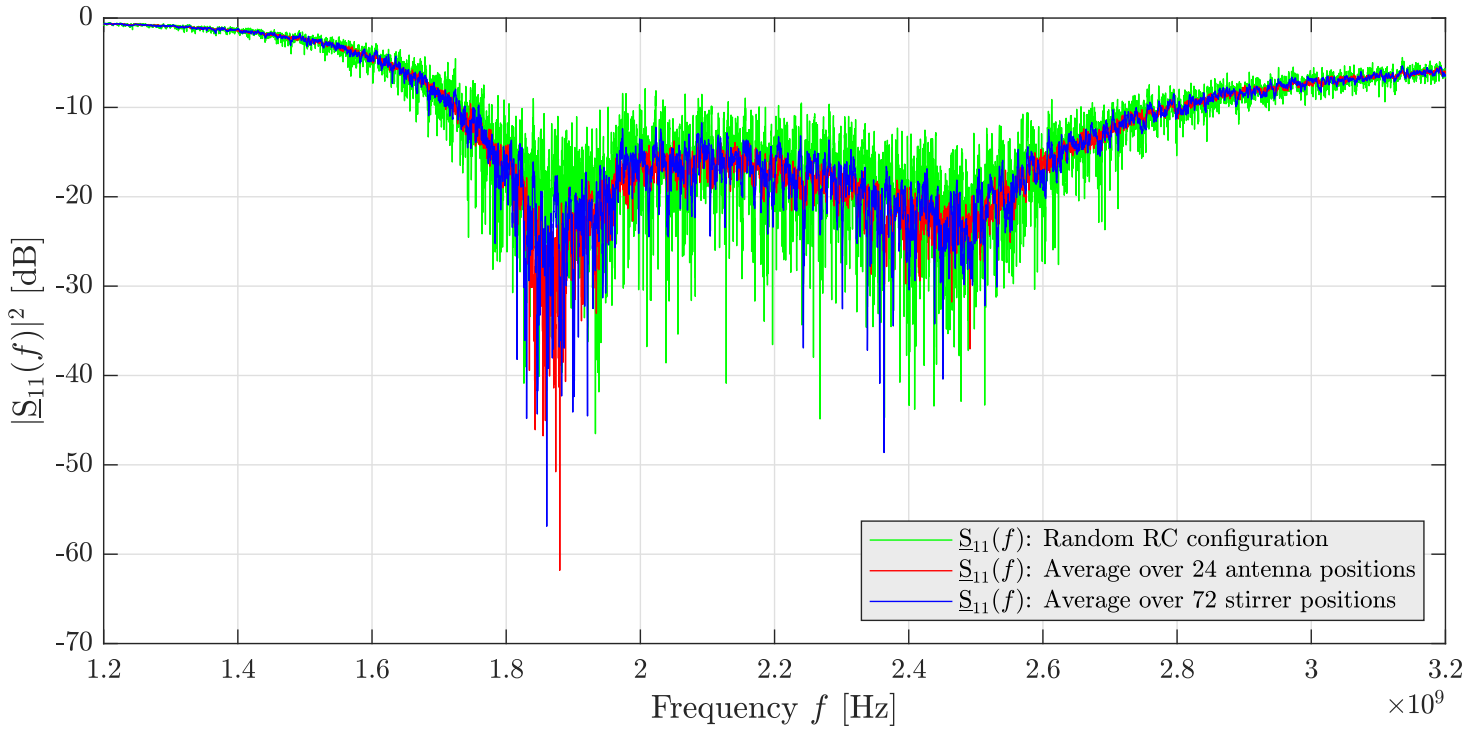


Figure 11: $\underline{S}_{11}(f)$: Reflection Coefficient of the AUT Measured in an RC

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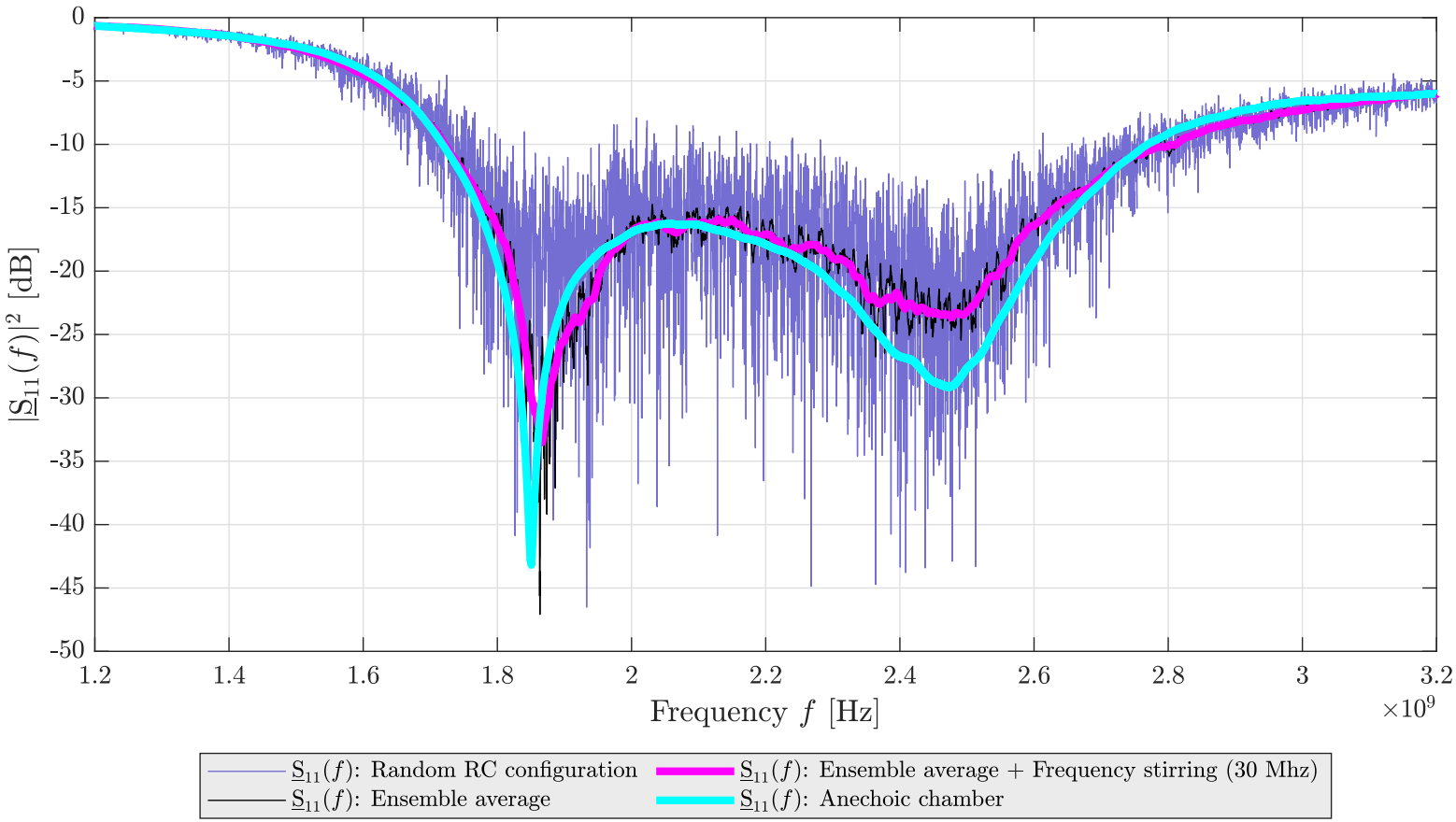


Figure 12: $\underline{S}_{11}(f)$: Different Types of Stirring Methods and Anechoic Chamber Measurement

Efficiency of stirring methods

Mean Squared Error

$$\mathrm{MSE} = \frac{1}{10001} \sum_{k=1}^{10001} \left(\underline{\mathrm{S}}_{11}(f_k) - \hat{\underline{\mathrm{S}}}_{11}(f_k) \right)^2$$

$$\mathrm{MSE}(\text{Ensemble average}) = 1.998 \times 10^{-4} \quad \mathrm{MSE}(\text{Mechanical stirring}) = 0.6667$$

$$\mathrm{MSE}(\text{Source stirring}) = 1.5 \times 10^{-3}$$

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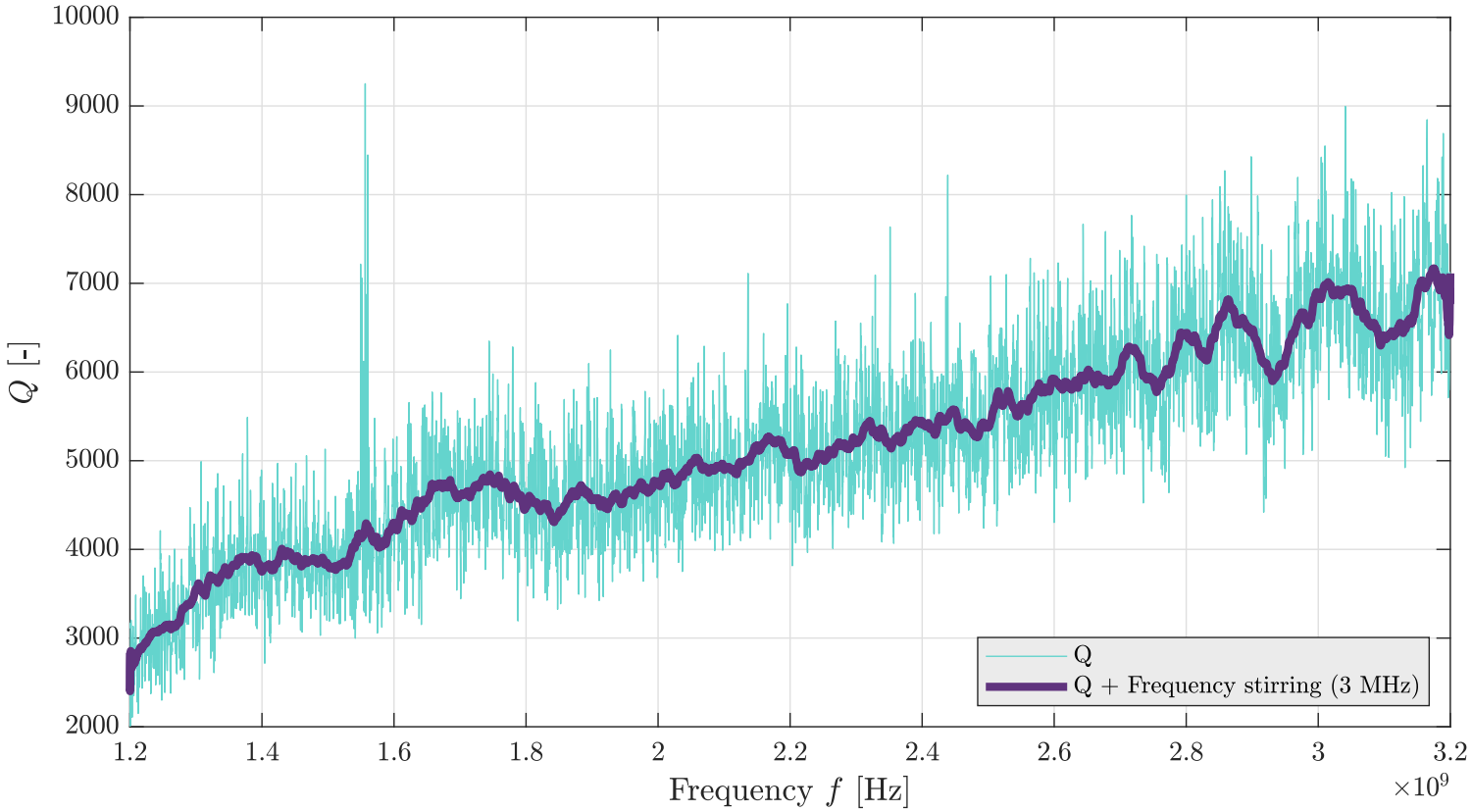


Figure 13: Q-factor of the Reverberation Chamber

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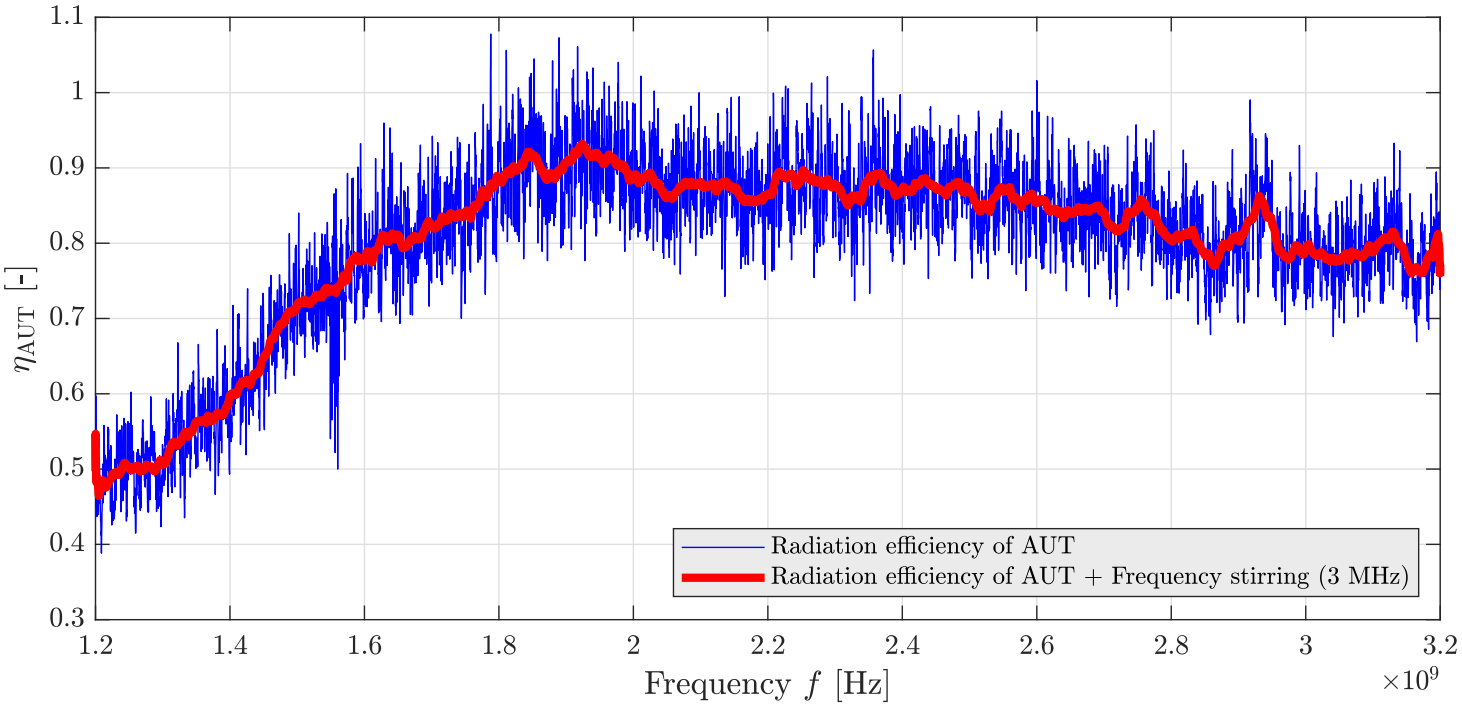


Figure 14: Radiation Efficiency of the Miniature Patch Antenna (AUT)

Data Processing and Antenna Radiation Efficiency Establishment (Holloway) [6]

Difference of value for time-domain estimation and frequency-domain estimation

$$Q_{\mathrm{TD}} = \omega\tau_{\mathrm{RC}} \neq Q_{\mathrm{FD}} = \frac{16\pi^2 V}{\lambda^3}\langle|\underline{S}_{21}(f)|^2\rangle$$

Obtaining $\frac{1}{\tau_{\mathrm{RC}}}$: slope of $\ln(\mathrm{PDP}(t)) = \ln\left(\left\langle\left|\mathcal{F}^{-1}[\underline{S}_{21\,\mathrm{sampled}}(f)]\right|^2\right\rangle\right)$

Retrieving Radiation Efficiency

$$\eta_1 = \sqrt{\frac{4\pi V}{f\lambda^3}\frac{\frac{\langle|\underline{S}_{11,\mathrm{Stirred}}(f)|^2\rangle}{\left(1-\langle|\underline{S}_{11}(f)|^2\rangle\right)^2}}{\tau_{\mathrm{RC}}}}$$

Data Processing and Antenna Radiation Efficiency Establishment (Holloway) [6]

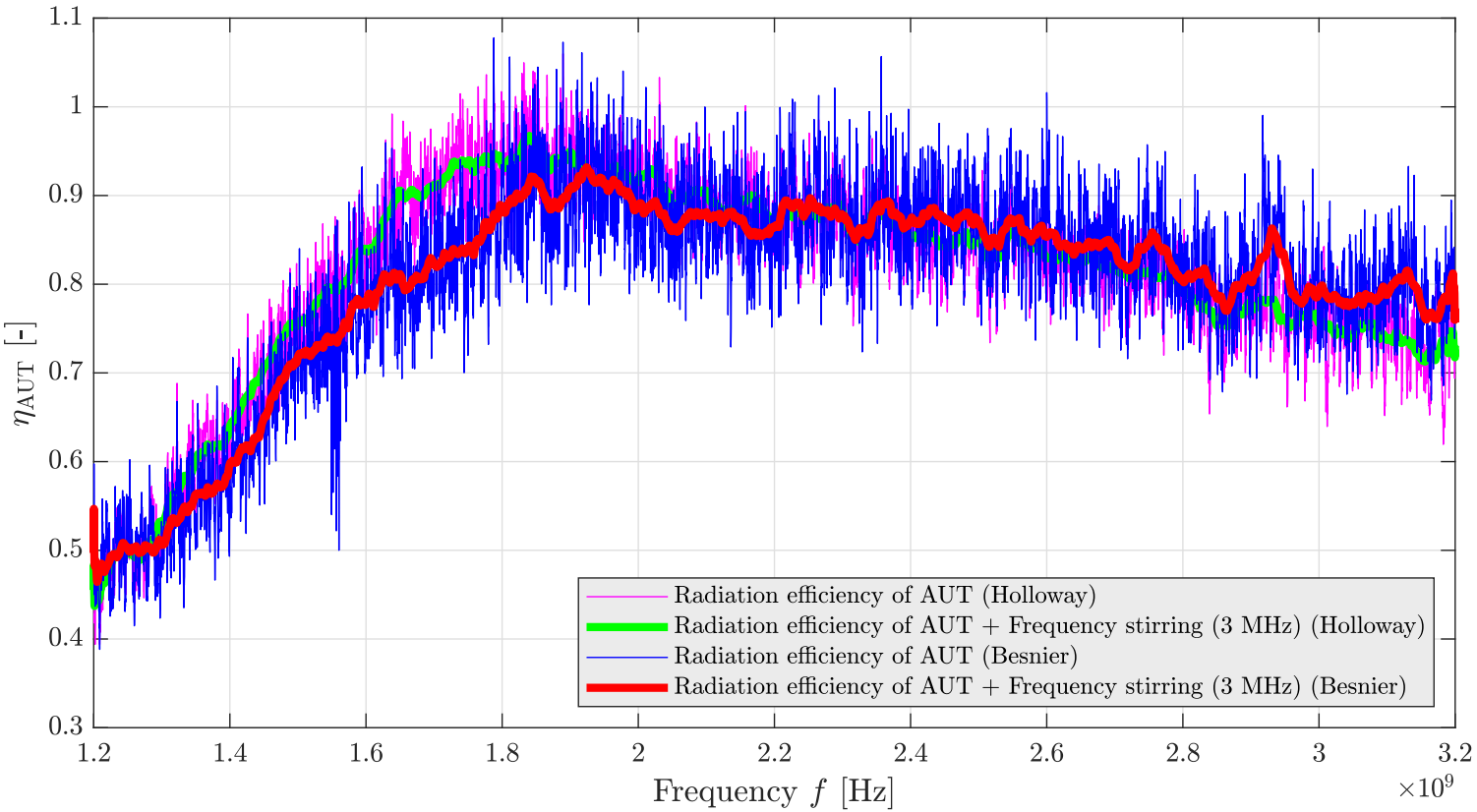


Figure 15: Radiation Efficiency of the Miniature Patch Antenna (AUT) Besnier Versus Holloway

Conclusion


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22/02/2021: Sharing theoretical parts between the two partners. Vincent: Antennas and S-parameters section Jesús: Reverberation Chamber and Q-method.

Reused Knowledge and Skills

- Electromagnetism
- Electronics
- Statistical Physics
- Acoustic Waves
- Signal Processing
- Radiating Transfer
- English

Developed Knowledge and Skills

- Radiation and Scattering Parameters
- Matlab
- LaTeX
- Autonomy
- Communication

Reused Knowledge and Skills

- Electromagnetism
- Electronics
- Wave propagation phenomenon
- Mechanics
- Matlab
- Statistics and Probability
- Fourier Transform
- English

Vincent Poitier

Developed Knowledge and Skills

- LaTeX
- GeoGebra
- TikZ
- Matlab
- Transmission line, Radiation of Antennas, Scattering parameters and Characteristics of Antennas
- English